EOJIN KIM

Audio Signal Processing & Deep Learning

Research Potfolio · 2026



EOJIN KIM

AUDIO SIGNAL PROCESSING
& DEEP LEARNING

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github

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RESEARCH INTERESTS

- Audio Signal Processing
- SELD
- Acoustic Localization
- Deep Learning

PROFILE

M.S. candidate specializing in audio signal processing and deep learning, with research experience spanning single-source localization in simulated noisy environments to multi-source sound event localization and detection (SELD) using real-world recordings.

I focus on identifying the root causes of performance limitations from both input features and model architectures, and on designing principled AI systems grounded in signal-processing fundamentals and practical deployment considerations.

EDUCATION & EXPERIENCE

2025 – 2027 **M.S. in Information and Communication Engineering**

Soongsil University, Seoul, Korea | GPA: 4.50 / 4.50

2019 – 2025 **B.S. in Electronic Engineering**

Soongsil University, Seoul, Korea
Relevant Coursework : Semiconductor Devices, Semiconductor & Display Processing, Analog Layout Design

2020 – 2022 **Republic of Korea Air Force**

17th Fighter Wing | Wired Communications Maintenance

RESEARCH EXPERIENCE

2025 – Present **Transmission Compensation Laboratory**

Graduate Researcher | Soongsil University

Audio signal processing · Acoustic localization · Deep learning

SELECTED RESEARCH PROJECTS

2026 – Present **Factory Facility Anomaly Detection and Localization**

Researcher | Disaster acoustic localization

Microphone-domain SELD · FAAF · ResNet-Conformer / CST-Former

2024 – 2025 **Abnormal Sound-Voice Detection in Disaster Environments**

Researcher | Disaster acoustic localization

Robust sound localization · GCC-PHAT · Acoustic simulation

SELECTED PUBLICATIONS

1 **Frequency-Adaptive Auditory Frontend for Microphone-Domain SELD**

E. Kim, S. Im · IEEE MLSP 2026

2 **Robust Sound Source Localization Using CDR-Based Masking and CNN-Enhanced GCC-PHAT**

E. Kim et al. · ICEIC 2025

TECHNICAL SKILLS

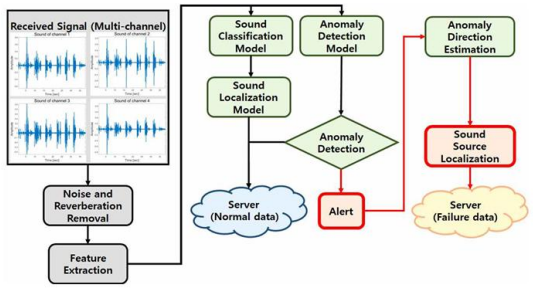
Python · PyTorch | GCC-PHAT · TDOA · STFT | SELD · Pyroomacoustics · Cadence Virtuoso

SELECTED RESEARCH

Frequency-Adaptive Auditory Frontend for Microphone-Domain SELD

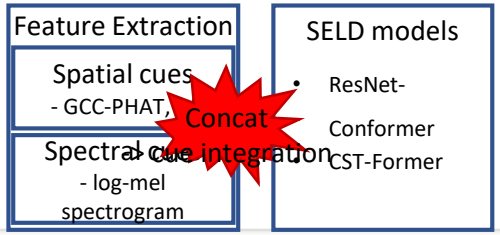
IEEE MLSP 2026 · First Author · Auditory-inspired feature integration for robust SELD

01 RESEARCH CONTEXT



- Industrial audio requires both event detection and source localization.
- Noise and reverberation make spatial estimation unreliable.

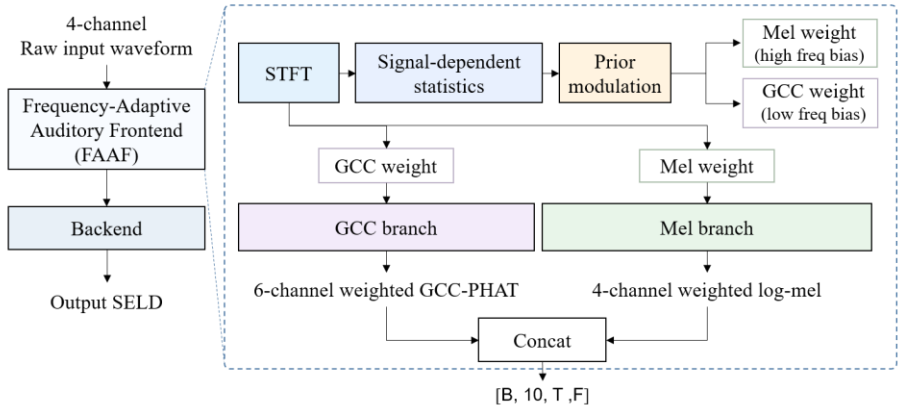
02 CORE LIMITATION



Conventional MIC-domain SELD leaves cue integration largely to the backend.

03 PROPOSED METHOD

FAAF inserts a frequency-aware frontend before the SELD backbone.



MY CONTRIBUTIONS

Auditory prior

Duplex-theory-inspired frequency bias

Signal-dependent modulation

Input statistics adapt the cue balance

Plug-in frontend

Works before different SELD backends

FAAF learns which acoustic cue to emphasize at each frequency — before the backend.

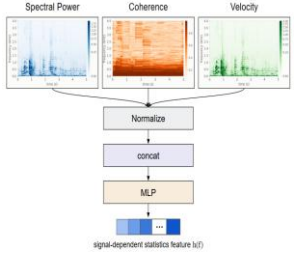
RESULTS & ANALYSIS

What FAAF learns — and why it matters

A three-step view of the frontend and quantitative validation across two SELD backends

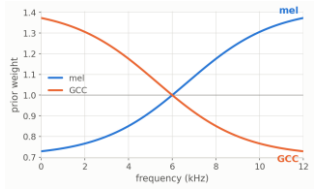
1 Analyze the input

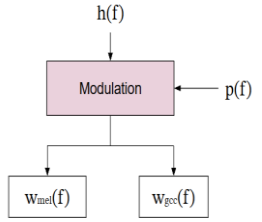
Power · coherence · velocity



2 Modulate the auditory prior

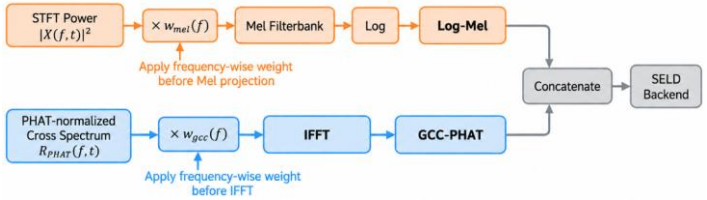
Signal statistics × prior





3 Reweight spectral & spatial cues

Frequency-wise mel / GCC weighting



04 QUANTITATIVE RESULTS

SELD score ↓ lower is better

ResNet-Conformer

0.4228 → 0.4069

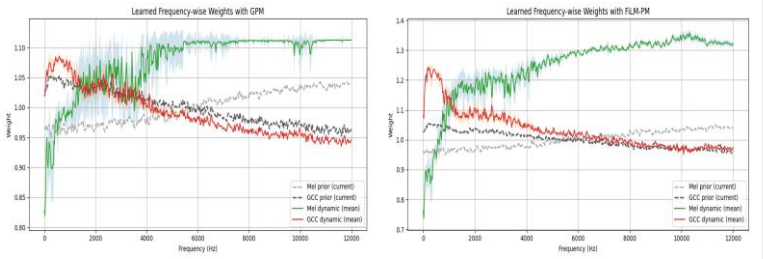
−3.8% relative

CST-Former

0.4800 → 0.4693

−2.2% relative

05 LEARNED BEHAVIOR



- Lower frequencies place more emphasis on spatial GCC cues.
- Higher frequencies increasingly favor spectral mel cues.
- The trend is qualitatively consistent with Duplex Theory.

TAKEAWAY FAAF improves two different SELD backends while learning an interpretable, frequency-dependent cue allocation pattern.

SELECTED RESEARCH

Robust Sound Source Localization Using CDR-Based Masking and CNN-Enhanced GCC-PHAT

ICEIC 2025 · First Author · Robust localization for noisy and reverberant disaster environments

01 RESEARCH CONTEXT



- Disaster response still relies heavily on human-centered search and rescue.
- Visual sensing degrades under debris, smoke, dust, and poor visibility.

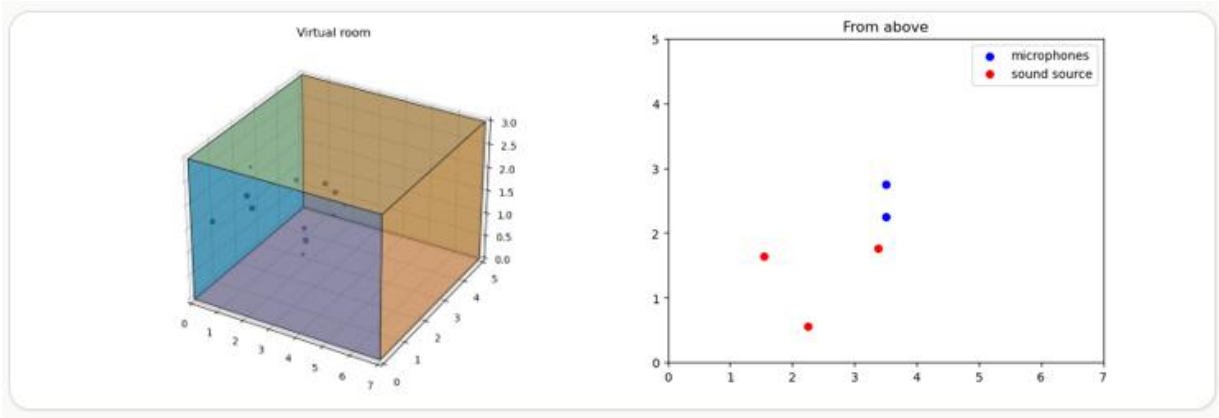
02 CORE LIMITATION

Real disaster-scene acoustic data are scarce and difficult to collect under controlled conditions.

Noise and reverberation further distort TDOA estimation.

03 CONTROLLED ACOUSTIC SIMULATION

Built a Pyroomacoustics-based simulation pipeline to generate acoustic data under controlled room, source, and microphone conditions.



Random source positions

Spatial diversity

Variable microphone geometry

Array variation

Noise & reverberation control

Condition-specific evaluation

TAKEAWAY A controlled simulation pipeline enabled systematic evaluation before moving toward real-world disaster acoustics.

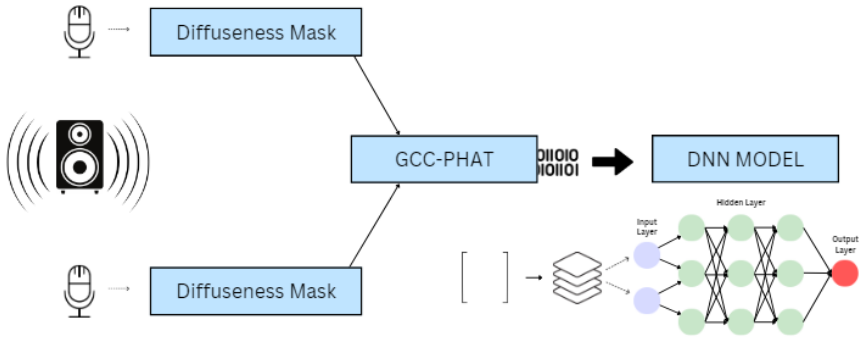
METHOD & RESULTS

Robust TDOA Estimation under Noise and Reverberation

CDR-based masking → parameterized GCC-PHAT → CNN-based TDOA estimation

01 PROPOSED METHOD

A compact front-end suppresses diffuse noise before extracting correlation features for learning-based TDOA estimation.



CDR-based diffuseness masking → Parameterized GCC-PHAT → CNN-based TDOA estimation

02 KEY RESULT

T60 = 0.6 s · SNR = 0 dB

1.40 → 0.94

MAE (cm)

≈ 33% lower error than conventional GCC-PHAT

GCC-PHAT	1.40
PGCC	1.01
Ours	**0.94**

03 WHAT THIS SHOWS

Robustness
Lower TDOA error under severe noise and reverberation.

Across conditions
Stable gains were observed across multiple acoustic settings.

Research outcome
First-author work presented at ICEIC 2025.

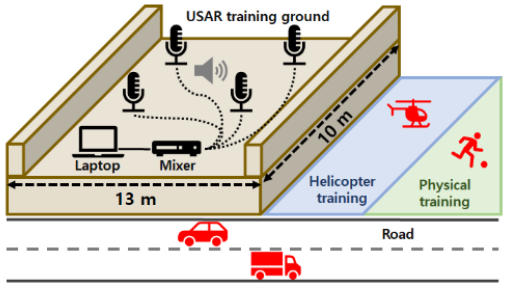
TAKEAWAY Physics-informed noise suppression and correlation features improved TDOA robustness without relying on a larger backbone.

FIELD VALIDATION

Real-World Acoustic Data Collection

Field Experiment at Gyeonggi Fire Service Academy

01 RECORDING ENVIRONMENT



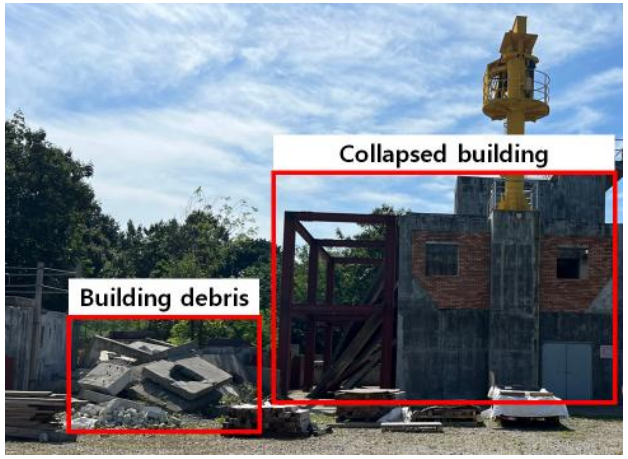
- 4-channel microphone array were used for sound source localization
- The training site included strong real-world interference.

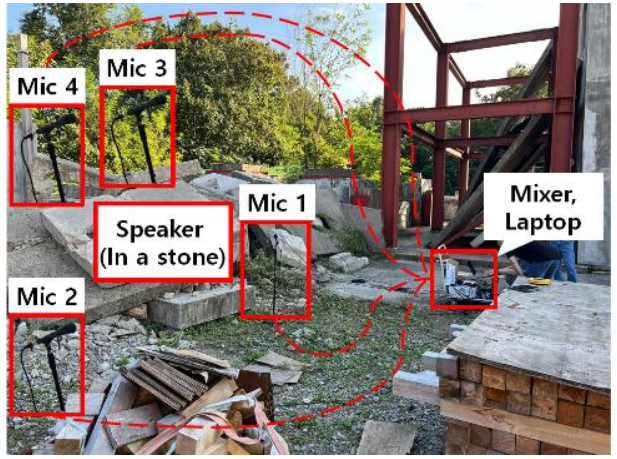
02 MY CONTRIBUTIONS

- Participated in microphone and loudspeaker placement
- Relocated the source across predefined coordinates during repeated recordings
- Logged source coordinates for each recording trial as localization ground truth

03 REAL-WORLD DATA COLLECTION

USAR training facility and field recording setup





Data collection site

Gyeonggi Fire Service Academy (GFSA), USAR Training Facility

Site characteristics

Collapsed-building environment, Challenging acoustic conditions

Practical significance

Evaluation of robustness & generalization under real-world acoustic conditions

TAKEAWAY Contributed to real-world multichannel acoustic data collection with spatial ground-truth annotations in a challenging USAR environment.

THANK YOU